

unident_s seed 0: baseline vs custom dense reward (ss=1.2)

Static report figure: cumulative deliveries and representative frames from the deterministic evaluation replay.

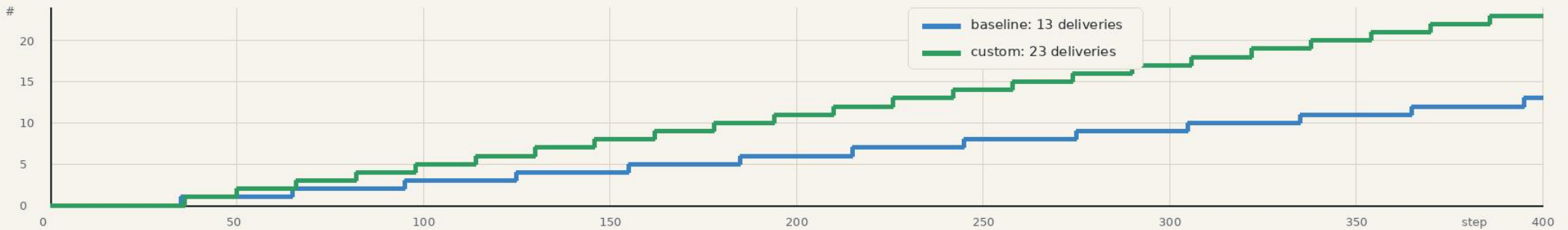
Baseline

13 deliveries; after first delivery, one soup every ~30 steps
delivery steps: [35, 65, 95, 125, 155] ... [335, 365, 395]

Custom ss=1.2

23 deliveries; after first delivery, one soup every ~16 steps
delivery steps: [36, 50, 66, 82, 98] ... [354, 370, 386]

Cumulative deliveries over 400-step episode



first soup (t=36)

Baseline



deliveries by t: 1
Custom ss=1.2



deliveries by t: 1

early steady cycle (t=130)

Baseline



deliveries by t: 4



deliveries by t: 7

mid episode (t=258)

Baseline



deliveries by t: 8



deliveries by t: 15

late episode (t=386)

Baseline



deliveries by t: 12
mean interval: 30.4 steps



deliveries by t: 23
mean interval: 16.8 steps

Interpretation

The custom-shaped policy does not mainly improve the first soup; both start delivering at about step 35-36.

Its gain comes from a shorter repeated production cycle: baseline settles into a conservative ~30-step loop, while custom ss=1.2 sustains ~16-step cycles.

This suggests the shaped reward makes intermediate progress states easier to reinforce once the agents discover a workable role pattern.